BENNETT UNIVERSITY | SCHOOL OF COMPUTER SCIENCE ENGINEERING & TECHNOLOGY

CSET400C — Capstone Project Progress Review Meeting (Venue: Room M255 | 21 October 2026, 11:00 AM)

Project Status Briefing: Elythra — AI Resume Screening & Ranking System

Course Code:	CSET400C (Capstone Project)	**Meeting Date / Time:**	21 Oct 2026, 11:00 AM (M255)
Team Leader:	Amandeep Verma (B.Tech CSE)	**Faculty Coordinator:**	Dr. Jabir Ali (Associate Professor)
Current Phase:	Post-Milestone 1 -> Milestone 2 Prototyping	**Backend Test Suite:**	test_pipeline.py (100% PASS)

1. Modular Development Status (Done vs. In Progress vs. Next)

Pipeline Stage	Target Modules & Files	Implementation Status	Verification Evidence
1. Document Ingestion	core/parser.py, core/text_cleaner.py	**COMPLETED**	Parses PDF, DOCX, TXT; normalizes UTF-8; handles encoding errors.
2. Data Sanitization & Ethics	core/text_cleaner.py	**COMPLETED**	Strips email, phone, and demographic markers to prevent evaluator bias.
3. Skill Taxonomy & Extraction	core/skills_database.py, core/extractor.py	**COMPLETED**	500+ skills categorized into 10 domains (Cloud, AI/ML, Languages, DBs...).
4. Multi-Factor Scoring Engine	core/matcher.py	**IN PROGRESS (Active)**	Tuning weights: 45% Skill Overlap + 35% TF-IDF + 10% Exp + 10% Edu.
5. Web UI & Visualization	app.py, utils/visualizer.py, utils/exporter.py	**PROTOTYPING (Milestone 2)**	Streamlit candidate leaderboard, Plotly gauges, CSV/Excel reporting.

2. Mathematical Formulation & Scoring Matrix

To ensure complete transparency and avoid black-box AI opacity, Elythra calculates match percentages using an explicit 4-factor formula:

Composite Match Score = (0.45 × S_overlap) + (0.35 × Sim_tfidf) + (0.10 × E_exp) + (0.10 × E_edu)

- **S_overlap (45%):** Jaccard-like intersection of candidate extracted skills against Job Description required skills.
- **Sim_tfidf (35%):** Sublinear TF-IDF vectorization with (1,2) n-grams and Cosine Similarity measuring domain contextual relevance.
- **E_exp (10%):** Ratio of candidate years of experience against JD minimum requirements (capped at 100%).
- **E_edu (10%):** Hierarchical degree alignment score (e.g. B.Tech / M.Tech / BCA / MCA matching minimum requirements).

3. Engineering Challenges Addressed So Far

- **Bias Mitigation:** Personal contact information (emails, phone numbers, external portfolio URLs) and college pedigrees are stripped by text_cleaner.py before the text is vectorized, ensuring evaluations are merit-based.
- **Handling Formatting Scramble:** Multi-column PDF resumes often cause text lines to merge awkwardly. We implemented regex sanitization and dual-pass parsing to preserve semantic boundaries.
- **Preventing Keyword Stuffing:** Because TF-IDF text similarity and experience verification are weighted alongside raw skill counts, applicants cannot game the system simply by repeating buzzwords.

4. Deliverables in Progress for Milestone 2 (MVP Submission)

- 1. Complete fine-tuning of the TF-IDF sigmoid scaling parameters across diverse non-technical and technical roles.
- 2. Finalize the Streamlit recruiter dashboard layout, adding interactive score threshold sliders and candidate deep-dive inspection.
- 3. Implement automated CSV and multi-tab Excel export functionality for recruitment audits.
- 4. Prepare formal test cases and validation results comparing Elythra ranking outputs against manual recruiter benchmarks.

Submitted By: Amandeep Verma (B.Tech CSE)

Reviewed By: Dr. Jabir Ali (Course Coordinator)